

L-ascorbic acid exists in equilibrium between compounds **1** and **2** in acidic solution. What type of functional group is formed when Compound **1** cyclizes?

- ☐ A. A lactam group [12%]
- ☐ B. An amide group [1%]
- ✓ ☒ C. A lactone group [79%]
- ☐ D. An anhydride group [7%]

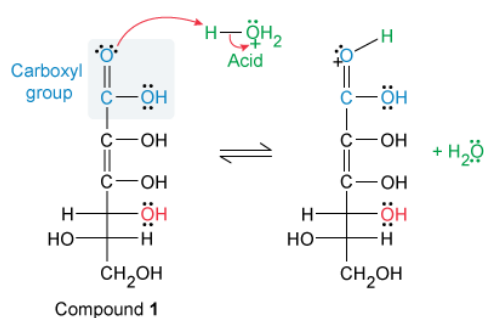
Correct

78% answered correctly

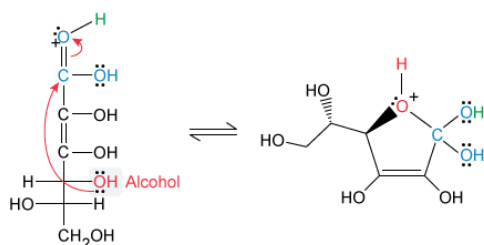
Explanation:

Functional groups are parts of a molecule that have different chemical properties. Under the right conditions, certain functional groups can react together to create a new functional group. **Lactones** are cyclic esters (**carboxylic acid derivative**) that form by an **intramolecular Fischer esterification** between a **carboxylic acid** and **alcohol** within the same molecule. They are **named** based on the number of carbons in the ring (excluding the carbonyl).

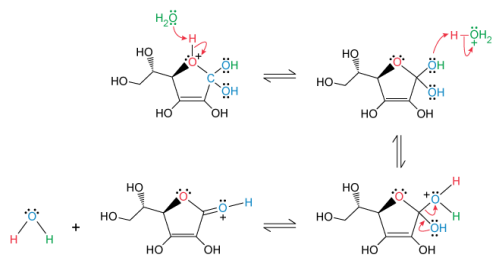
Lactone formation begins with protonation of the carboxyl group by an acid (H_3O^+).



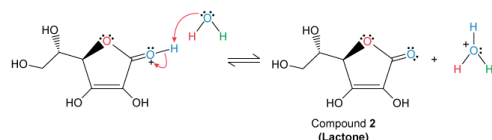
The alcohol hydroxyl group then nucleophilically attacks the carbonyl to form a five-membered ring.



A proton is then transferred from the alcohol hydroxyl group to the carboxylic acid hydroxyl group, resulting in water (a **good leaving group**).



Finally, the carbonyl oxygen is deprotonated.



Compound **1** in this question contains both a carboxylic acid and an alcohol. Under acidic conditions, an intramolecular Fischer esterification occurs in which compound **1** cyclizes, forming compound **2**, a cyclic molecule containing an oxygen atom adjacent to a carbonyl within the ring. Therefore, a **lactone forms** when compound **1** cyclizes.

(Choice A) A **lactam** is a cyclic amide rather than a cyclic *ester*.

(Choice B) An **amide** is a carboxylic acid derivative with a *nitrogen* rather than an oxygen adjacent to a carbonyl. Amides are **formed** from a carboxylic acid and an *amine*.

(Choice D) An **anhydride** is a carboxylic acid derivative **formed** from a carboxylic acid and an *acid chloride*, or the condensation of *two carboxylic acids*.

Educational objective:

Lactones (cyclic esters) are carboxylic acid derivatives named for the number of carbons in the ring system (excluding the carbonyl). Lactones are formed by an acid catalyzed reaction between a carboxylic acid and an alcohol in the same molecule (ie, intramolecular Fischer esterification).

Time spent:0 sec

Subject:
Organic Chemistry

QID:403339

System:
Functional Groups and Their Reactions